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A

B

C

D

E

F

G

H

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01\_carrier\_interface

01\_carrier\_interface.kicad\_sch

10\_input\_protection

10\_input\_protection.kicad\_sch

20\_positive\_rails

20\_positive\_rails.kicad\_sch

30\_negative\_reference

30\_negative\_reference.kicad\_sch

40\_power\_supervision

40\_power\_supervision.kicad\_sch

03\_peripheral\_boundaries

03\_peripheral\_boundaries.kicad\_sch

04\_dac\_0\_3

04\_dac\_0\_3.kicad\_sch

05\_dac\_4\_7

05\_dac\_4\_7.kicad\_sch

06\_analog\_outputs

06\_analog\_outputs.kicad\_sch

41\_power\_status

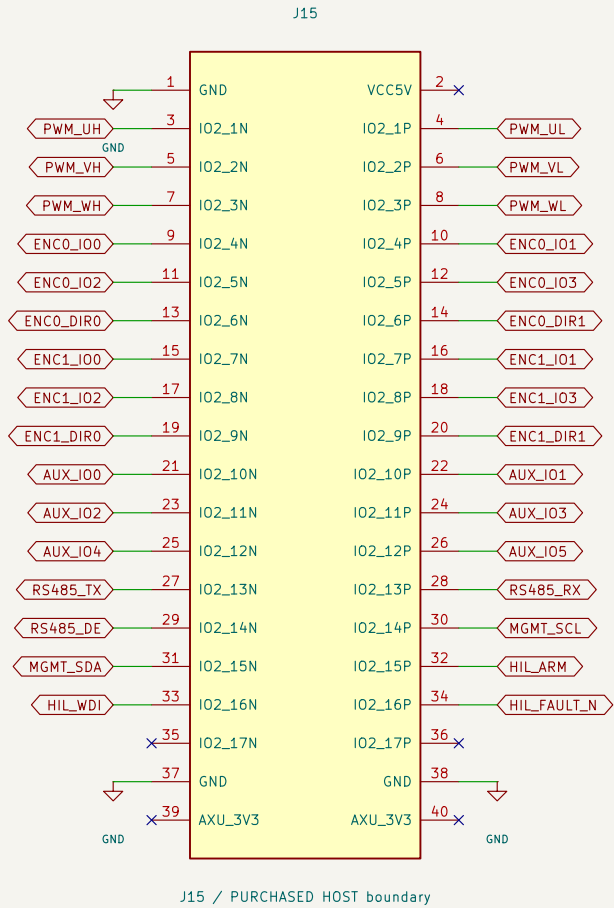
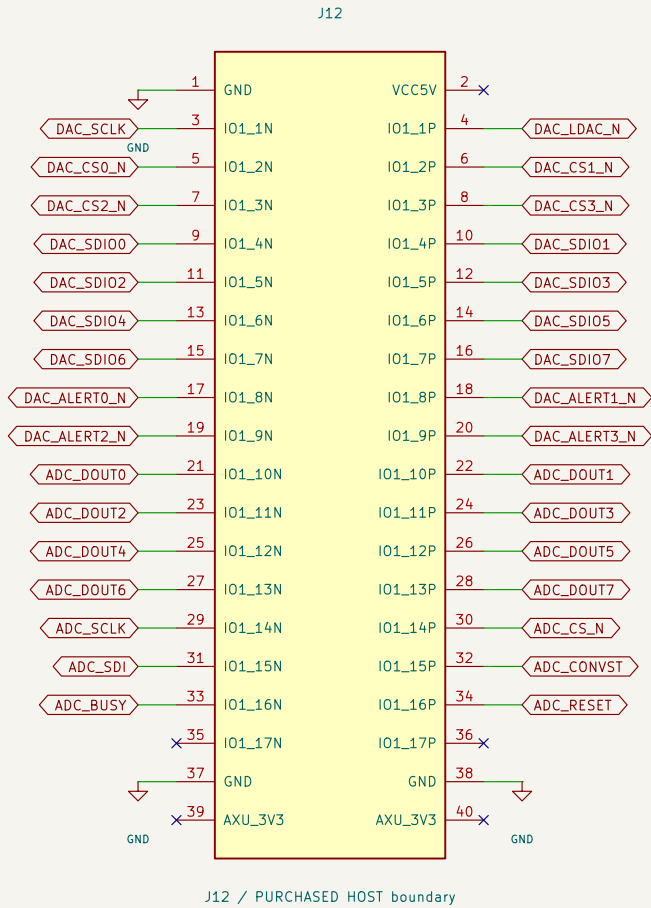
41\_power\_status.kicad\_sch

Sheet: /  
File: servohl\_io\_revB.kicad\_sch

Title: ServoHIL Rev.B – native editable single-SoC expansion

Size: A2Date:KICad E.D.A. 10.0.6

Rev: B–power–developmentId: 1/11



Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB-I or -E. No board-power backfeed through contacts.

Signal-pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

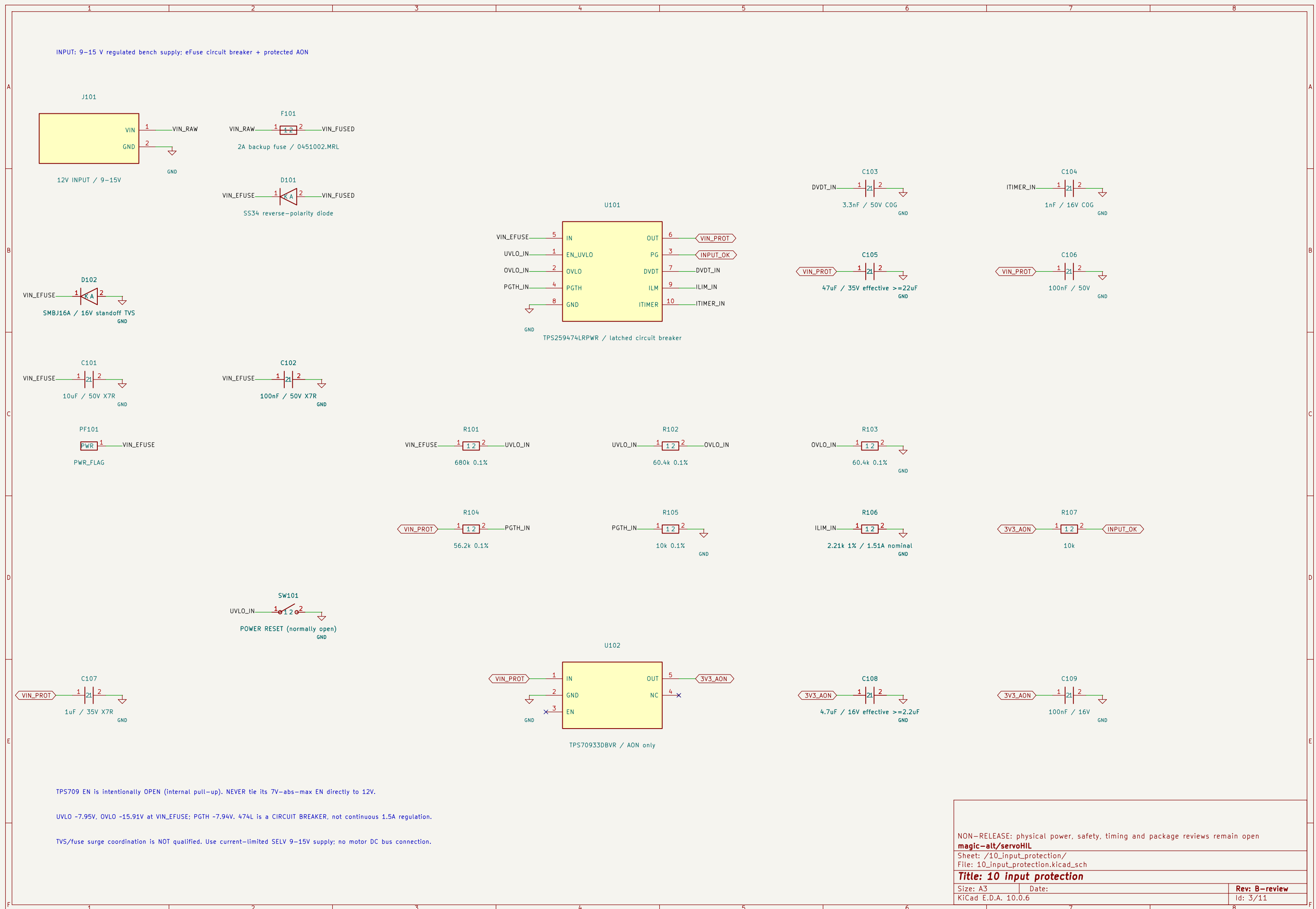
NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-att/servoHIL**

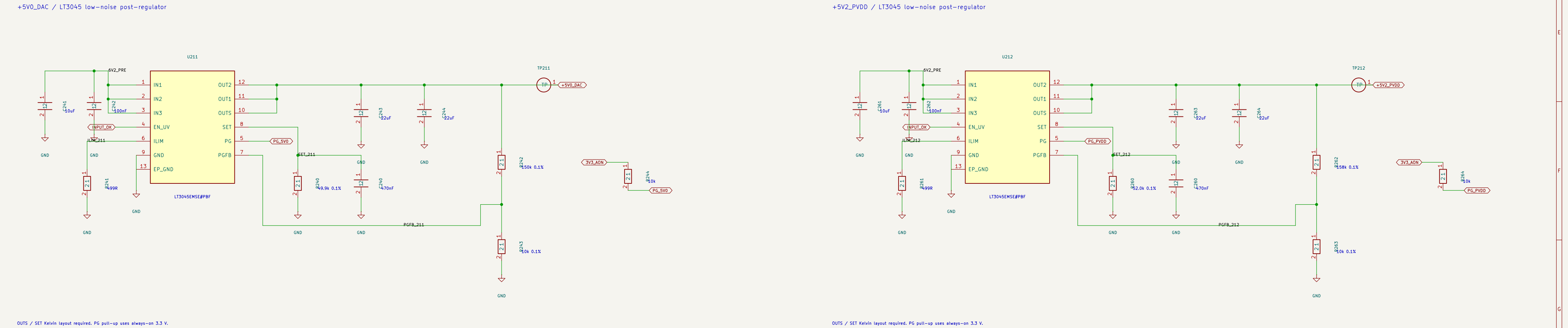
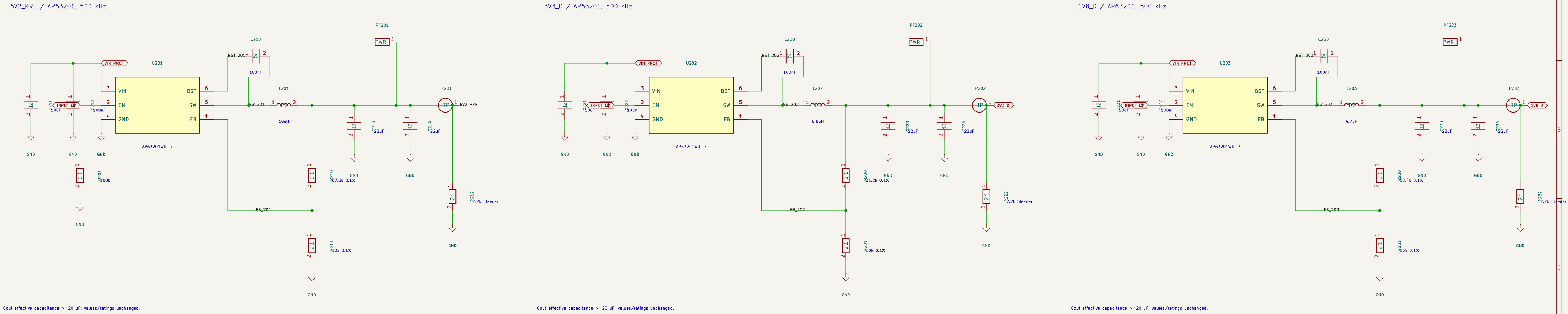
Sheet: /01\_carrier\_interface/  
File: 01\_carrier\_interface.kicad\_sch

**Title: 01 carrier interface**

|                     |       |               |
|---------------------|-------|---------------|
| Size: A3            | Date: | Rev: B-review |
| KiCad E.D.A. 10.0.6 |       | Id: 2/11      |



POSITIVE RAILS – direct power path, local feedback, cross-sheet interfaces only



All parts, pin numbers, values and footprints match the previous native circuit. These drawings do not establish bench qualification.

No second FPGA. Output current, capacitor DC bias, startup, ripple, thermal and safety behavior remain hardware review gates.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /20\_positive\_rails/

File: 20\_positive\_rails.kicad\_sch

**Title: 20 positive rails**

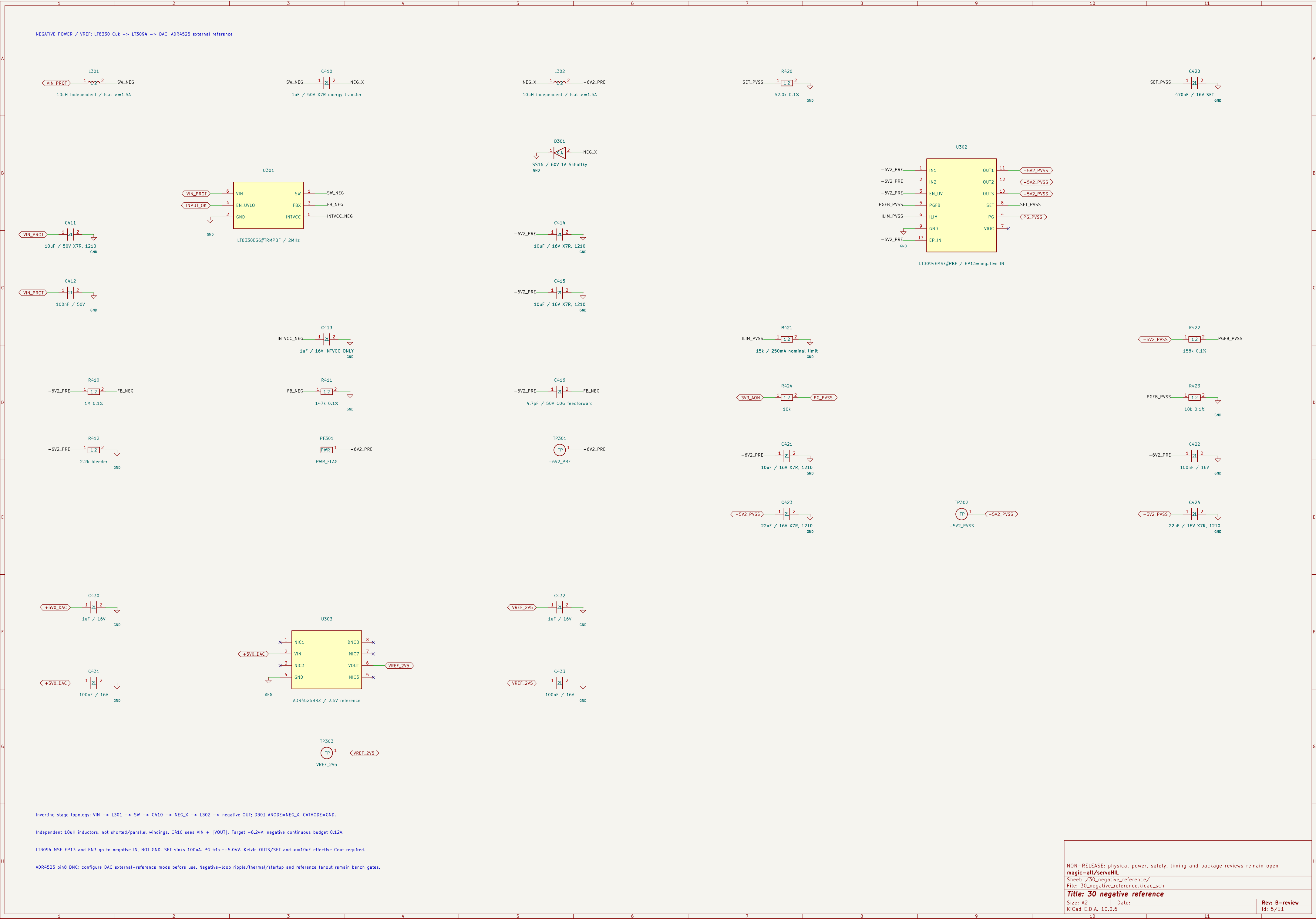
Size: A2

Date:

Rev: B-review

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Id: 4/11



NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /30\_negative\_reference/

File: 30\_negative\_reference.kicad\_sch

**Title: 30 negative reference**

Size: A2

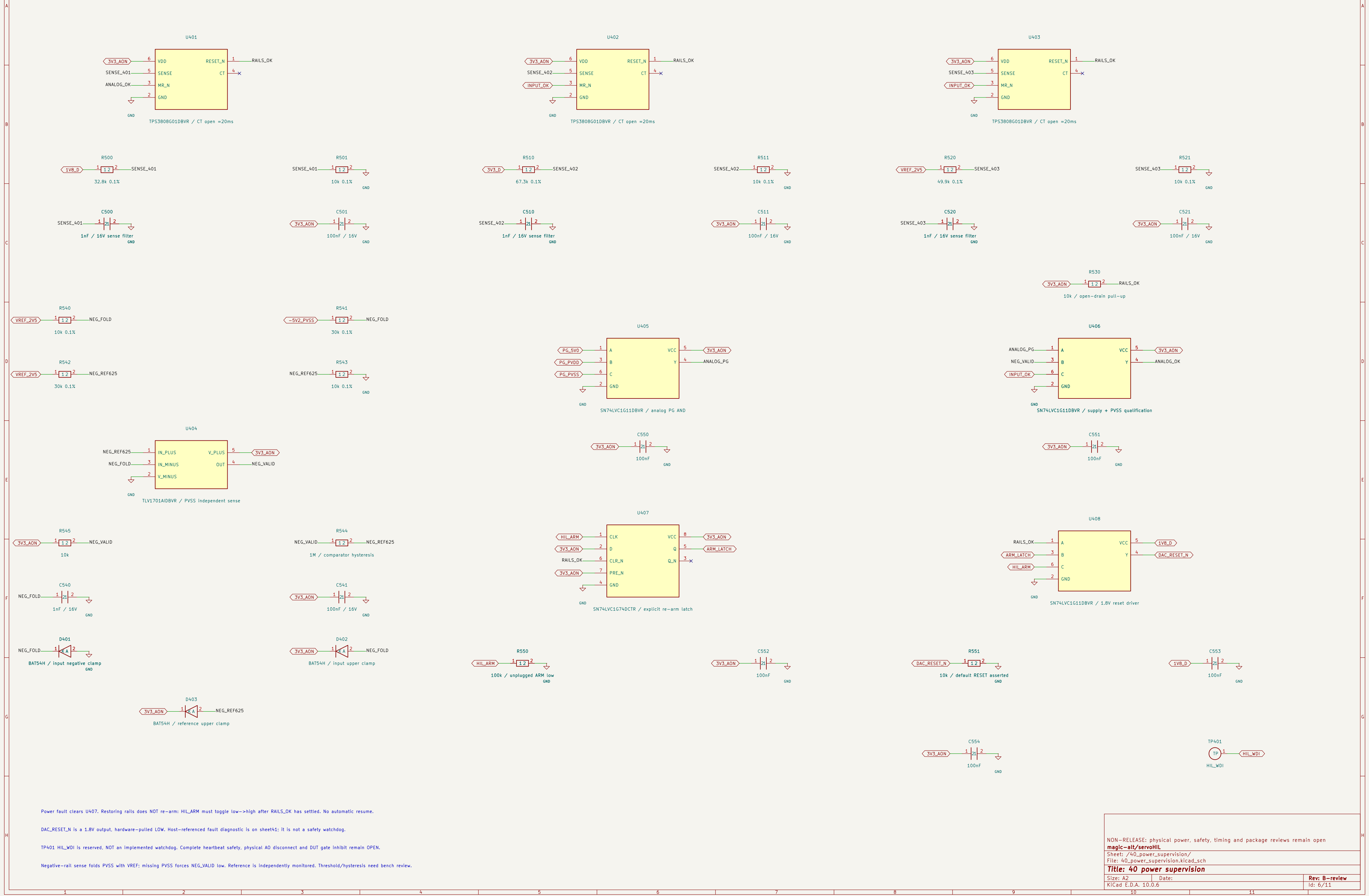
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 5/11

POWER QUALIFICATION: monitored rails, negative-rail loss detection and latched DAC reset release



Power fault clears U407. Restoring rails does NOT re-arm: HIL\_ARM must toggle low-->high after RAILS\_OK has settled. No automatic resume.

DAC\_RESET\_N is a 1.8V output, hardware-pulled LOW. Host-referenced fault diagnostic is on sheet41; it is not a safety watchdog.

TP401 HIL\_WDI is reserved, NOT an implemented watchdog. Complete heartbeat safety, physical AO disconnect and DUT gate Inhibit remain OPEN.

Negative-rail sense folds PVSS with VREF; missing PVSS forces NEG\_VALID low. Reference is independently monitored. Threshold/hysteresis need bench review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /40\_power\_supervision/

File: 40\_power\_supervision.kicad\_sch

**Title: 40 power supervision**

Size: A2

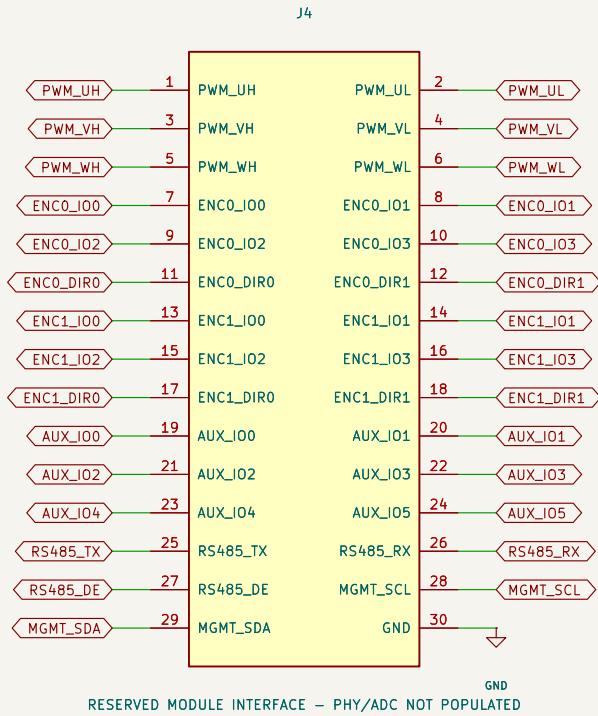
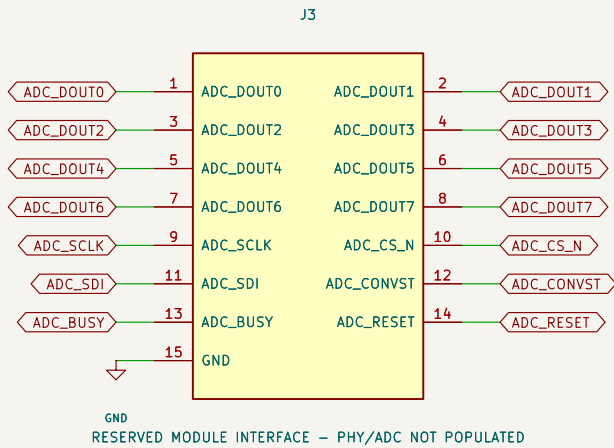
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 6/11

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

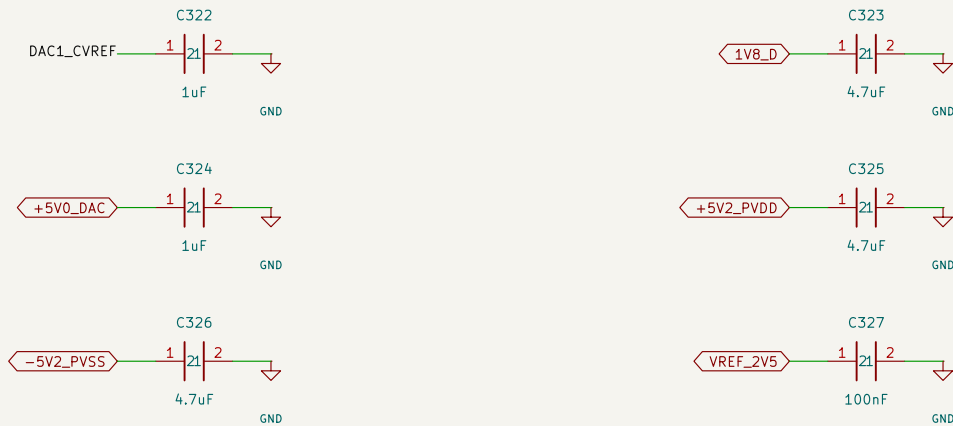
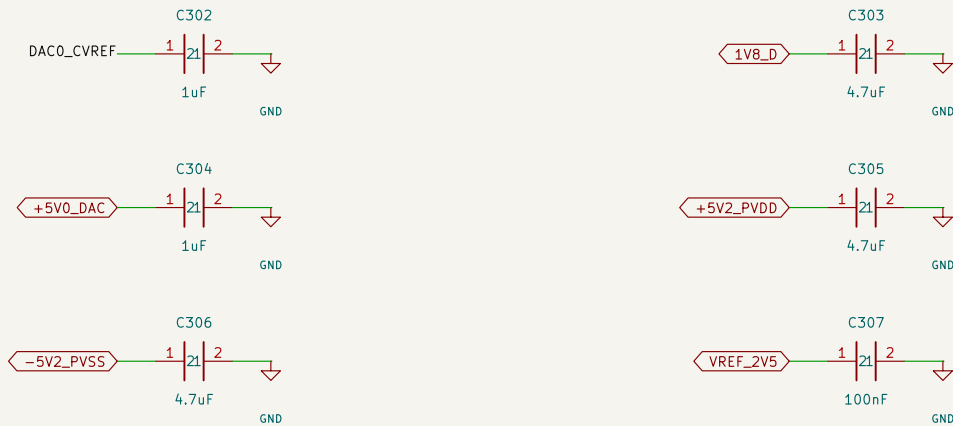
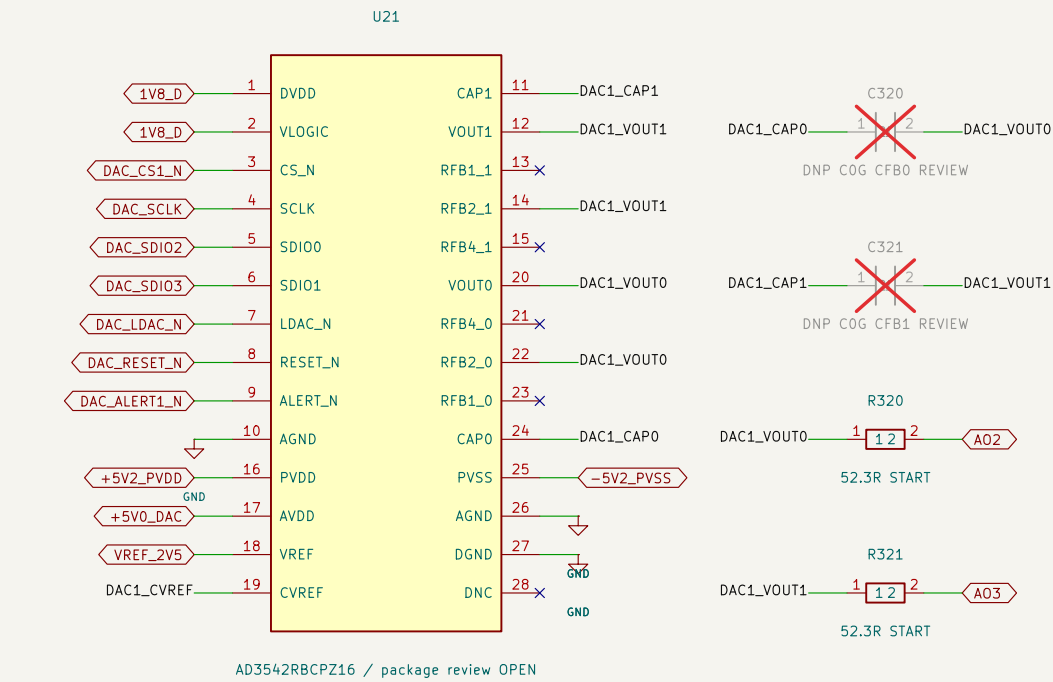
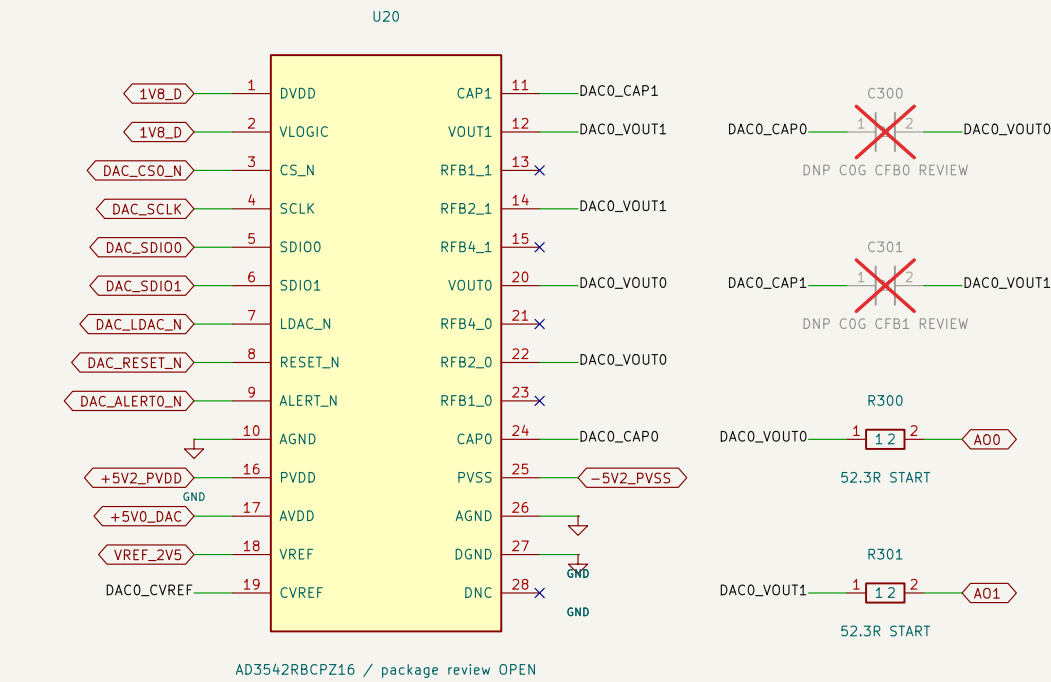
NON–RELEASE: physical power, safety, timing and package reviews remain open

**~~magic~~–alt/servoHIL**

Sheet: /03\_peripheral\_boundaries/  
File: 03\_peripheral\_boundaries.kicad\_sch

**Title: 03 peripheral boundaries**

|                     |       |               |
|---------------------|-------|---------------|
| Size: A3            | Date: | Rev: B–review |
| KiCad E.D.A. 10.0.6 |       | Id: 7/11      |



RFB2 selects +/-5V capability; RFB1/RFB4/DNC are not connected, CFB and load stability are NOT approved.

External reference configuration and power–up register settings must be verified before enabling outputs.

NON–RELEASE: physical power, safety, timing and package reviews remain open

**magic–alt/servoHIL**

Sheet: /04\_dac\_0\_3/

File: 04\_dac\_0\_3.kicad\_sch

**Title: 04 dac 0 3**

Size: A3

Date:

**Rev: B–review**

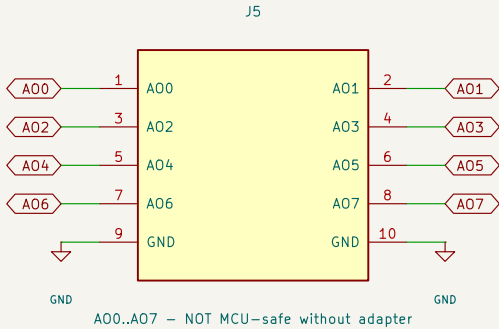
KiCad E.D.A. 10.0.6

Id: 8/11

External reference configuration and power-up register settings must be verified before enabling outputs.

|                     |       |               |
|---------------------|-------|---------------|
| Title: 05 dac 4 7   |       |               |
| Size: A3            | Date: | Rev: B-review |
| KiCad E.D.A. 10.0.6 |       | Id: 9/11      |

06 – Eight analog outputs to reviewed DUT adapter



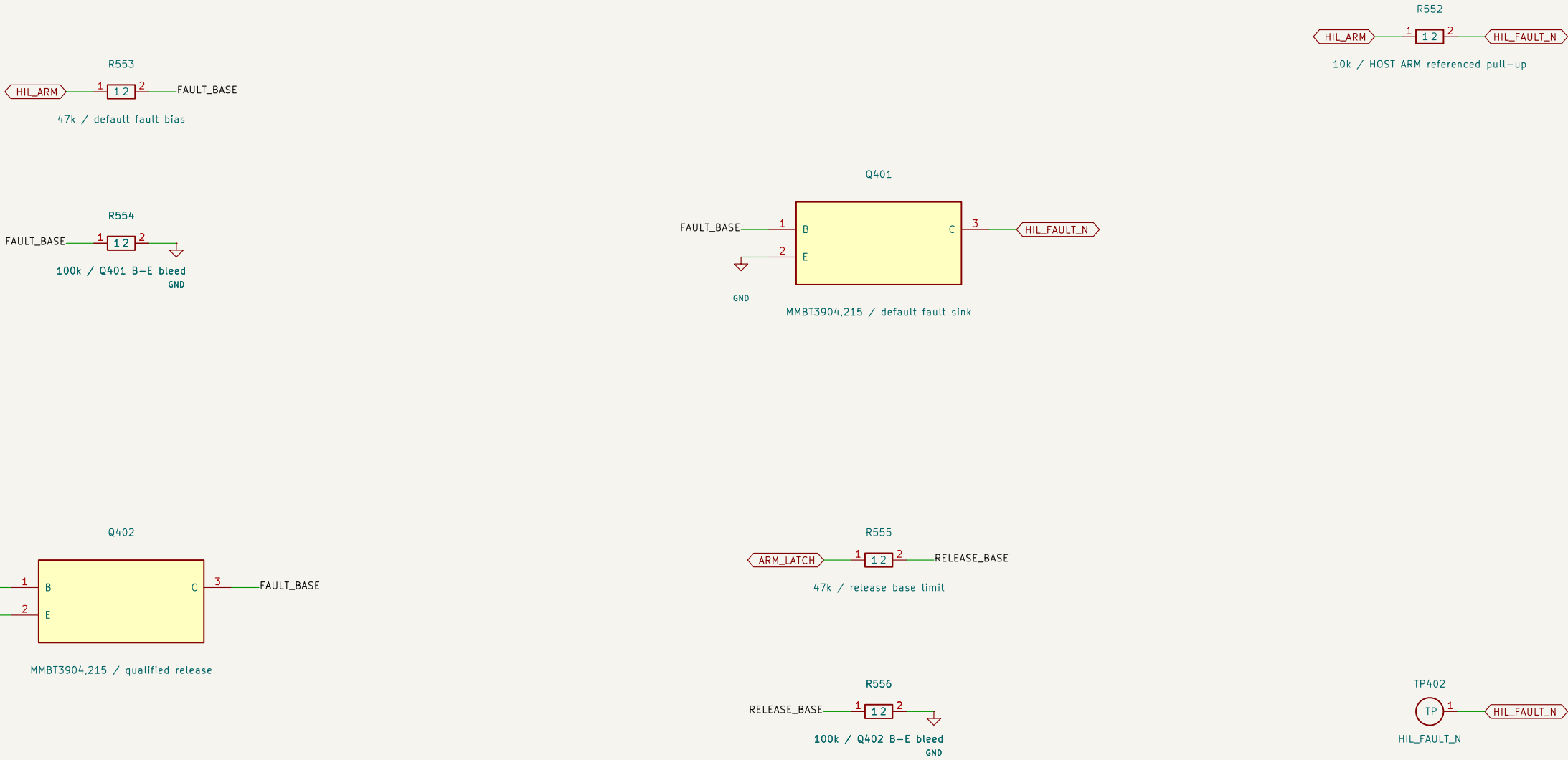
Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

NON-RELEASE: physical power, safety, timing and package reviews remain open  
**~~magic-alt/servoHIL~~**  
Sheet: /06\_analog\_outputs/  
File: 06\_analog\_outputs.kicad\_sch

|                          |       |               |
|--------------------------|-------|---------------|
| Title: 06 analog outputs |       |               |
| Size: A3                 | Date: | Rev: B-review |
| KiCad E.D.A. 10.0.6      |       | Id: 10/11     |

HOST-REFERENCED POWER STATUS: no expansion-rail pull-up on HIL\_FAULT\_N



HIL\_ARM is generated by the powered HOST, NOT an expansion rail. Q401 defaults ON while HOST requests ARM.

Only a power-qualified ARM\_LATCH turns Q402 on and releases Q401. Board power lost -> Q402 off -> Q401 reports fault.

HIL\_FAULT\_N=HIGH means power-qualified AND armed; LOW also means disarmed. At HOST off, no expansion supply drives this pin high.

Nominal fault sink <0.33mA and HOST ARM total load <0.4mA. Check transient/partial-power behavior physically; not certified safety.

Other direct DAC/ADC pins STILL require power-off backfeed protection. This diagnostic alone does not authorize hot plugging.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /41\_power\_status/

File: 41\_power\_status.kicad\_sch

**Title: 41 power status**

Size: A3

Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 11/11